Dense nonperiodic limit-periodic potentials with finite interval spectra

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Abstract

We prove that real nonperiodic limit-periodic potentials whose spectrum is a finite union of nondegenerate intervals are dense among all real limit-periodic potentials. Finite block insertions fill limiting spectral clusters, while separately protected marker blocks ensure nonperiodicity. This answers AIM 1031 affirmatively.

Repository target. This manuscript concerns AIM problem 1031 at revision `3fa50884d129`. Its scope is stated below.

The answer to Problem 1031 is **yes**. In fact, potentials whose entire spectrum is a finite union of nondegenerate closed intervals are dense among limit-periodic potentials, and the approximating potentials can always be required to be nonperiodic. The proof places finite-support approximate eigenvectors inside long periodic blocks. Small constant shifts of those blocks fill successively smaller spectral gaps, while separate marker blocks ensure that the limiting potential is nonperiodic.

Theorem

Let $V : \mathbb{Z} \to \mathbb{R}$ be a bounded uniform limit of periodic sequences, and let $\varepsilon > 0$. Then there exists a nonperiodic uniform limit of periodic sequences W such that

$$\|W - V\|_\infty < \varepsilon, \qquad \sigma(H_W) = \bigcup_{j=1}^{M} [\ell_j, r_j], \qquad \ell_j < r_j < \ell_{j+1},$$

where

$$(H_W u)(n) = u(n-1) + u(n+1) + W(n)u(n) \quad \text{on } \ell^2(\mathbb{Z}).$$

The interval union is finite, with $M \geq 1$, and the displayed separation condition applies for $j < M$. In particular, the spectrum has nonempty interior and is not a Cantor set. The initial potential need not itself be nonperiodic. Thus the theorem has all the quantifiers required in the catalogue question, with a stronger conclusion. If V is constant, one may take $M = 1$.

1. A finite block insertion lemma

Let P be a bounded real potential of integer period q. Fix a number $a > 0$, a positive integer K, and a finite list of target energies $E_1, \dots, E_m$. Suppose that

$$\operatorname{dist}(E_j, \sigma(H_P)) \le a \qquad (1 \le j \le m).$$

For every $\eta > 0$, there is a periodic potential $\widetilde{P}$, whose period is a multiple of q, with the following three properties:

$$\|\widetilde{P} - P\|_\infty \le a, \tag{1}$$

$$\operatorname{dist}(E_j, \sigma(H_{\widetilde{P}})) < \eta \qquad (1 \le j \le m), \tag{2}$$

and, for every integer $1 \le p \le K$, there is an integer n_p such that

$$|\widetilde{P}(n_p + p) - \widetilde{P}(n_p)| \ge 2a. \tag{3}$$

Proof of the lemma

Choose $e_j \in \sigma(H_P)$ with $|E_j - e_j| \le a$, and set $c_j = E_j - e_j$. Since adding a constant to a potential shifts its spectrum by that constant,

$$E_j \in \sigma(H_{P+c_j}).$$

For a bounded self-adjoint operator, every spectral point is an approximate eigenvalue. Finitely supported sequences are dense in $\ell^2(\mathbb{Z})$, and the operator is bounded, so there is a finitely supported unit vector ϕ_j with

$$\|(H_{P+c_j} - E_j)\phi_j\|_2 < \eta. \tag{4}$$

Choose an integer block length L, divisible by q, which exceeds K and is sufficiently large to accommodate each support after translation by a multiple of q. On a finite sequence of consecutive blocks of length L, prescribe the potential as follows:

$$P + c_1,\ P + c_2,\ \dots,\ P + c_m,\ P + a,\ P - a,\ P + a. \tag{5}$$

Then repeat this entire block pattern periodically. Because every block boundary is a multiple of q, the background P has the same phase on every block. The resulting potential $\widetilde{P}$ has period $(m + 3)L$, a multiple of q, and satisfies (1).

Translate ϕ_j by a multiple of q so that its support lies in the block carrying $P + c_j$. On that support the two potentials agree exactly. Their difference is a multiplication operator, so the translated vector still has the residual in (4), even though the potentials outside the block may differ. The spectral theorem gives

$$\operatorname{dist}(E_j, \sigma(H_{\widetilde{P}})) \le \|(H_{\widetilde{P}} - E_j)\phi_j^{\text{translated}}\|_2 < \eta.$$

This proves (2).

For (3), use the two internal boundaries in the three marker blocks $P + a, P - a, P + a$. Call them b_1, b_2; both are multiples of q. For $1 \le p \le K < L$, the pair of sites $b_i - 1, b_i + p - 1$ crosses

the corresponding boundary and stays within its two adjacent marker blocks. In both cases the background difference is the same number

$$v_p = P(p-1) - P(-1).$$

The two differences of the new potential are therefore

$$v_p - 2a, \qquad v_p + 2a.$$

At least one has absolute value at least $2a$. This proves (3) simultaneously for every $p \le K$. □

2. Spectral clusters and the perturbation budget

Choose a periodic potential V_0 with

$$\|V - V_0\|_\infty < \frac{\varepsilon}{2}.$$

Write its spectrum as its finitely many connected components:

$$S_0 = \sigma(H_{V_0}) = \bigcup_{j=1}^{M} I_{0,j}, \qquad I_{0,j} = [\ell_{0,j}, r_{0,j}], \qquad \ell_{0,j} < r_{0,j} < \ell_{0,j+1}. \tag{6}$$

The intervals are nondegenerate by the elementary Floquet description of the spectrum of a periodic discrete Schrödinger operator. Touching bands are combined into a single component. If $M > 1$, let

$$G = \min_{j<M}(\ell_{0,j+1} - r_{0,j}) > 0.$$

Choose a number t satisfying

$$0 < t < \varepsilon, \qquad t < G/4 \quad \text{if } M > 1.$$

For $k \ge 1$, set

$$a_k = t4^{-k}, \qquad \delta_k = \frac{a_{k+1}}{8} = \frac{a_k}{32}, \qquad \delta_0 = 0. \tag{7}$$

Thus

$$\sum_{k\ge1} a_k = \frac{t}{3}, \qquad \sum_{j>k} a_j = \frac{a_k}{3}, \qquad \delta_{k-1} = \frac{a_k}{8} \quad (k \ge 2). \tag{8}$$

The initial gaps will remain open throughout the construction. Indeed, each endpoint can move by at most $\sum a_k = t/3$, so all gaps between the original M clusters remain at least $G - 2t/3 > G/2$ when $M > 1$.

We shall construct periodic potentials V_k, with periods divisible by those of their predecessors, and ordered disjoint intervals $I_{k,1}, \dots, I_{k,M}$ such that, with $S_k = \sigma(H_{V_k})$,

$$\|V_k - V_{k-1}\|_\infty \le a_k, \tag{9}$$

$$S_k \subset \bigcup_{j=1}^{M} I_{k,j}, \qquad \sup_{E \in \bigcup_j I_{k,j}} \operatorname{dist}(E, S_k) \le \delta_k, \tag{10}$$

and each $I_{k,j}$ is the convex hull of the part of S_k in its corresponding cluster. Its endpoints therefore belong to S_k. We will also retain the marker condition

$$\forall\, 1 \le p \le k \quad \exists\, n_{k,p} \in \mathbb{Z} : \quad |V_k(n_{k,p}+p) - V_k(n_{k,p})| \ge 2a_k. \tag{11}$$

At $k = 0$, (10) holds with $\delta_0 = 0$ by (6).

3. Filling each expanded cluster

Suppose V_{k-1} and its cluster hulls have been constructed. Define the expanded intervals

$$K_{k,j} = I_{k-1,j} + [-a_k, a_k], \qquad K_k = \bigcup_{j=1}^{M} K_{k,j}. \tag{12}$$

The mesh bound in (10), together with $\delta_{k-1} < a_k$, gives the exact identity

$$S_{k-1} + [-a_k, a_k] = K_k. \tag{13}$$

For completeness, a point inside a previous hull is within δ_{k-1} of S_{k-1}; a point in one of its two extensions is within a_k of the corresponding endpoint, which belongs to S_{k-1}. This proves one inclusion. The reverse inclusion follows from $S_{k-1} \subset \bigcup_j I_{k-1,j}$. The case $k = 1$ uses $\delta_0 = 0$.

Choose a finite $\delta_k/4$-net $E_1, \dots, E_m$ of the entire compact set K_k. By (13), every target satisfies

$$\operatorname{dist}(E_j, S_{k-1}) \le a_k.$$

Apply the finite block insertion lemma with $P = V_{k-1}$, $a = a_k$, $K = k$, and $\eta = \delta_k/4$. It gives a periodic potential V_k satisfying (9), (11), and

$$\sup_{E \in K_k} \operatorname{dist}(E, S_k) \le \delta_k. \tag{14}$$

The residual estimate actually gives $\delta_k/2$, but the weaker bound is convenient.

For bounded self-adjoint operators, spectral Hausdorff distance is at most operator-norm distance. Since the difference of the Schrödinger operators is multiplication by the difference of their potentials, (9) and (13) also give the exact confinement

$$S_k \subset S_{k-1} + [-a_k, a_k] = K_k. \tag{15}$$

This is what controls the entire spectrum, including spectral points associated with interfaces between the blocks.

Define

$$I_{k,j} = \operatorname{conv}(S_k \cap K_{k,j}) = [\ell_{k,j}, r_{k,j}]. \tag{16}$$

The sets being convexified are nonempty. If there is only one cluster, this follows from (14). If $M > 1$, the gaps between the $K_{k,j}$ exceed $G/2$, while $\delta_k \le t/128 < G/512$; hence the spectral

points within δ_k of either endpoint of $K_{k,j}$ must lie in that same cluster. Compactness then gives the endpoint bounds

$$\ell_{k-1,j} - a_k \le \ell_{k,j} \le \ell_{k-1,j} - a_k + \delta_k, \tag{17}$$

$$r_{k-1,j} + a_k - \delta_k \le r_{k,j} \le r_{k-1,j} + a_k. \tag{18}$$

In particular,

$$I_{k-1,j} \subset I_{k,j} \subset K_{k,j}. \tag{19}$$

These inclusions also verify inductively the claimed lower bound on gaps between different clusters. Because $I_{k,j} \subset K_{k,j}$, (14) proves the mesh bound (10); the confinement (15) proves its other inclusion. The induction is complete.

The tolerance choices are not circular. The current spectral mesh is fixed by increasing block lengths, without increasing the allowed perturbation amplitude. The next step then uses $\delta_k = a_{k+1}/8$ to ensure (13).

4. The limiting spectrum is exactly a finite interval union

The summable perturbation bounds imply that V_k converges uniformly to a bounded real limit-periodic potential W. Moreover,

$$\|W - V\|_\infty \le \|W - V_0\|_\infty + \|V_0 - V\|_\infty < \frac{t}{3} + \frac{\varepsilon}{2} < \varepsilon, \tag{20}$$

and

$$\|W - V_k\|_\infty \le \frac{a_k}{3}. \tag{21}$$

By (17)–(19), each sequence of left endpoints is decreasing and bounded, and each sequence of right endpoints is increasing and bounded. Write their limits as ℓ_j, r_j. The limiting intervals are nondegenerate because they contain $I_{0,j}$. Distinct intervals remain separated by at least $G - 2t/3 > 0$. Thus

$$K_\infty = \bigcup_{j=1}^{M} [\ell_j, r_j] \tag{22}$$

is a finite union of exactly M disjoint nondegenerate closed intervals. The endpoint movement after stage k is at most $\sum_{s>k} a_s = a_k/3$, so

$$d_H\left(\bigcup_j I_{k,j}, K_\infty\right) \le \frac{a_k}{3}. \tag{23}$$

The mesh and confinement in (10) say

$$d_H\left(S_k, \bigcup_j I_{k,j}\right) \le \delta_k. \tag{24}$$

At the same time, norm continuity of the spectrum and (21) give

$$d_H(S_k, \sigma(H_W)) \leq \frac{a_k}{3}. \tag{25}$$

Combining (23)–(25) and letting $k \to \infty$ yields

$$\boxed{\sigma(H_W) = K_\infty.} \tag{26}$$

This proves the asserted conclusion for the entire spectrum. In particular, if the initial periodic potential V_0 is constant, its spectrum is one interval and the construction retains $M = 1$.

5. The limit is nonperiodic

Fix an integer $p \geq 1$, and choose $k \geq p$. The marker property (11) gives a site $n_{k,p}$ where the p-shift difference of V_k has magnitude at least $2a_k$. The future tail cannot erase that difference:

$$\begin{aligned} |W(n_{k,p}+p) - W(n_{k,p})| &\geq 2a_k - 2\|W - V_k\|_\infty \\ &\geq 2a_k - \frac{2a_k}{3} = \frac{4a_k}{3} > 0. \end{aligned}$$

Therefore p is not a period of W. Since p was arbitrary, W is nonperiodic. Equations (20) and (26) now prove the theorem.

Scope and references

The construction guarantees that the entire spectrum is a finite union of nondegenerate closed intervals. It can be a single interval when the initial periodic approximant has connected spectrum; this includes approximation of any constant potential. The argument makes no assertion about the spectral type, so it does not resolve the separate mixed-type or coupling-dependence questions.

1. D. Damanik and J. Fillman, *Spectral Properties of Limit-Periodic Operators*, arXiv:1802.05794v1 (2018), Question 10.4, printed p. 36; subsequently Chapter 15 in *Analysis and Geometry on Graphs and Manifolds* (2020). The question explicitly requires density among nonperiodic limit-periodic potentials. https://arxiv.org/pdf/1802.05794.
2. D. Damanik, Z. Gan, and H. Krüger, *Limit-periodic Schrödinger operators with a discontinuous Lyapunov exponent*, Journal of Functional Analysis **279** (2020), 108565. Its Theorem 1.3 concerns density of discontinuous Lyapunov exponents and is not itself the spectral-interior statement proved here. https://doi.org/10.1016/j.jfa.2020.108565.

The proof uses only elementary periodic approximation, the approximate-eigenvector criterion, and norm continuity of the spectrum for bounded self-adjoint operators. No bibliographic-priority claim is made.